

Software Requirements Specification Draft

SENG 31242 - System Design Project

Project - Animal Arc

Group – Logic Lords (Group 6)

SE/2022/021

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SE/2022/033

SE/2022/050

*Prepared for Animal Ark, Faculty of Veterinary Medicine and Animal Science, University of
Peradeniya*

1. Introduction

1.1 Purpose and Scope

Purpose: The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for the Animal Arc system. Animal Arc is designed as a digital animal welfare management platform for Animal Ark, the student-led animal welfare society of the Faculty of Veterinary Medicine and Animal Science, University of Peradeniya.

The main goal of the system is to improve how Animal Ark manages rescued animals, treatment records, emergency cases, medicine stock, donations, blood donor information, adoption activities, volunteer contributions, and public engagement.

Scope: The Animal Arc system will provide a centralized platform for volunteer veterinary students, supervisor doctors, society committee members, donors, adopters, and members of the public. The system will support both internal animal-care workflows and external public-facing interactions.

The core capabilities of the system include:

- **Animal Intake:** Allowing authorized volunteers to register rescued animals with rescue details, photos, location, condition, and initial notes.
- **Treatment Timeline:** Maintaining a chronological treatment history for each animal, including medicines, procedures, observations, and recovery progress.
- **Handover Notes:** Supporting shift-to-shift communication between volunteers by recording important notes and pending actions.
- **Supervisor Oversight:** Allowing supervisor doctors to review cases, approve treatment decisions, and monitor critical animals.
- **Case Closure:** Allowing animal cases to be closed after recovery, death, adoption, transfer, or release.
- **Emergency Alerts:** Notifying relevant volunteers and supervisors about urgent rescue or treatment situations.
- **Medicine Stock Management:** Tracking medicine inventory, stock levels, and low-stock situations.
- **Donation Management:** Recording donation campaigns, donor contributions, and donation usage.
- **Blood Donor Registration:** Maintaining a searchable registry of animal blood donors for emergency transfusion support.

- Adoption Flow: Managing adoption listings, adoption applications, screening, approval, and adoption completion.
- Volunteer Statistics: Recording volunteer involvement, case participation, and contribution statistics.
- Public Portal: Providing public users with access to adoption listings, donation campaigns, awareness posts, and contact information.

Out of Scope: The system will not include advanced hospital billing, online payment gateway integration, full veterinary hospital management, government animal registration, laboratory equipment integration, or AI-based diagnosis in the current version.

1.2 Definitions, Acronyms, Abbreviations

- Animal Ark: The student-led animal welfare society at the Faculty of Veterinary Medicine and Animal Science, University of Peradeniya.
- Animal Arc: The proposed software system developed to support Animal Ark rescue, treatment, adoption, donation, and volunteer management workflows.
- SRS: Software Requirements Specification.
- Volunteer: A veterinary student or authorized society member who participates in animal rescue, treatment support, handovers, and follow-up care.
- Supervisor Doctor: A veterinary doctor or authorized supervisor who reviews animal cases and provides treatment guidance.
- Committee Admin: A society committee member responsible for managing system records, campaigns, stock, adoption flow, and user access.
- Animal Intake: The process of registering a newly rescued animal into the system.
- Treatment Timeline: A chronological record of treatment updates, medicines, observations, and progress notes related to an animal.
- Handover Notes: Notes passed between volunteers during shift changes to ensure treatment continuity.
- Emergency Alert: A system notification used to inform responsible users about urgent rescue or treatment situations.
- Medicine Stock: The available inventory of medicines and medical supplies used by Animal Ark.
- Blood Donor: A registered animal that may be contacted for blood donation during emergency transfusion cases.
- Adopter: A member of the public who applies to adopt an animal listed by Animal Ark.

1.3 References

- Stakeholder Interview Summary 01
- Stakeholder Interview Summary 02
- Volunteer Survey Findings

- Existing WhatsApp and Paper-Based Workflow Analysis
- Existing System Analysis with Pain Points
- Use Case Descriptions in docs/srs/use-cases/
- Activity Diagrams in docs/srs/diagrams/
- SENG 31242 System Design Project Guidelines

1.4 Overview of Document

The remainder of this SRS document is structured as follows:

- Section 2: Overall Description provides a high-level overview of the system, user classes, operating environment, constraints, assumptions, and dependencies.
- Section 3: System Analysis describes the fact-finding techniques, existing system analysis, pain points, use cases, and diagram references.
- Section 4: Functional Requirements lists the specific functional requirements of the Animal Arc system with priority, source, and verification method.
- Section 5: Non-Functional Requirements defines the performance, security, usability, reliability, scalability, and maintainability requirements.
- Section 6: Alternative Solutions & Feasibility Study evaluates alternative approaches and justifies the recommended solution.

2. Overall Description

2.1 Product Perspective

Animal Arc is a web-based and mobile-responsive software system designed to support Animal Ark animal welfare operations. The system replaces the current fragmented workflow that depends heavily on WhatsApp messages, paper records, informal social media posts, and manually maintained donation or medicine records.

The system will act as a centralized platform where internal users can manage animal records, treatment updates, volunteer handovers, emergency alerts, donations, medicine stock, blood donors, and adoption processes. Public users will interact with the system through a public portal that displays adoption animals, donation campaigns, awareness content, and contact information.

The proposed system will contain the following main components:

- Internal dashboard for volunteers, supervisors, and committee admins.
- Animal profile and treatment management module.
- Emergency alert and notification module.

- Medicine stock management module.
- Donation and campaign management module.
- Blood donor registry module.
- Adoption management module.
- Volunteer statistics module.
- Public portal for donors, adopters, and general visitors.

2.2 User Classes and Characteristics

2.2.1 Volunteer Veterinary Student

Volunteer veterinary students are the primary operational users of the system. They participate in animal rescue, treatment updates, daily care, and handover communication. They require a simple and fast interface because they may use the system during busy clinical or rescue activities.

Main needs:

- Register rescued animals.
- Update treatment progress.
- Add handover notes.
- Respond to emergency alerts.
- View assigned or active animal cases.
- Track personal volunteer contribution statistics.

2.2.2 Supervisor Doctor

Supervisor doctors review animal cases, monitor critical animals, guide treatment decisions, and approve important case updates. They require accurate case histories and clear treatment timelines.

Main needs:

- Review treatment records.
- Provide medical instructions.
- Approve or reject case closure.
- Monitor emergency cases.
- View volunteer updates and handover notes.

2.2.3 Committee Admin

Committee admins manage society-level operations such as user access, medicine stock, donations, adoption applications, blood donor records, and public portal content.

Main needs:

- Manage user roles.
- Manage medicine inventory.
- Manage donation campaigns and records.
- Manage adoption listings and applications.
- View volunteer statistics.
- Maintain public portal content.

2.2.4 Donor

Donors are external users who support Animal Ark by contributing money, medicines, food, or other resources. They require clear campaign information and transparency about donation needs.

Main needs:

- View active donation campaigns.
- Submit donation interest or donation details.
- View basic information about how donations support animals.

2.2.5 Adopter

Adopters are public users who wish to adopt recovered animals. They require access to adoption listings and a clear application process.

Main needs:

- View animals available for adoption.
- Submit adoption applications.
- Provide contact and home environment details.
- Receive application status updates.

2.2.6 Public Visitor

Public visitors include animal lovers, students, university staff, and community members who want to learn about Animal Ark and support its activities.

Main needs:

- View public animal welfare information.
- View adoption posts and donation campaigns.
- Contact Animal Ark.
- Register as a potential blood donor owner if applicable.

2.3 Operating Environment

- Client Devices: Desktop computers, laptops, tablets, and smartphones.
- Web Browser Support: Chrome, Edge, Firefox, and Safari.
- Network Environment: The system assumes normal internet access for most operations. Some users may access the system using mobile data during rescue or treatment activities.
- Server Environment: The application will be hosted on a cloud-based or university-supported server environment.
- Database: A centralized database will store animal profiles, treatment records, user accounts, donation records, medicine stock records, adoption applications, and blood donor details.
- Media Storage: Animal photos, medical images, adoption photos, and donation proof images will be stored securely.

2.4 Design and Implementation Constraints

- The system must be simple enough for volunteer students to use during busy rescue and treatment situations.
- The system must support role-based access because volunteers, supervisors, admins, donors, adopters, and public users require different permissions.
- Sensitive information such as donor contact details, adopter information, and internal treatment records must not be publicly exposed.
- The public portal must not allow unauthorized users to modify internal records.
- The system must be designed within the SENG 31242 project timeline.
- The design must remain realistic for later implementation during the development phase.
- The system must support image uploads for animal profiles, treatment updates, adoption listings, and donation proof where required.

2.5 Assumptions and Dependencies

- It is assumed that Animal Ark volunteers and committee members have access to smartphones or computers with internet connectivity.
- It is assumed that supervisor doctors will review important medical updates through the system when required.
- It is assumed that the society will define authorized users before system deployment.
- It is assumed that adoption approval decisions will still be made by authorized Animal Ark members, not automatically by the system.
- It is assumed that donation verification may require manual confirmation by committee admins.
- It is assumed that medicine stock updates depend on accurate entry by authorized users.
- The system depends on a stable hosting environment, database service, and file storage service.

- The system depends on accurate stakeholder input gathered through interviews, surveys, and existing workflow analysis.

3. System Analysis

3.1 Fact-Gathering Techniques Used

3.1.1 Structured Interviews

Structured interviews were conducted with key stakeholders to understand Animal Ark current workflow, problems, and expectations. The interviews focused on animal intake, treatment tracking, emergency handling, adoption, donations, medicine stock, and volunteer coordination.

3.1.2 Volunteer Survey

A survey questionnaire was prepared for volunteer veterinary students to collect structured feedback about current issues in treatment tracking, handover communication, emergency alerts, medicine stock awareness, adoption processes, and volunteer recognition.

3.1.3 Existing System Analysis

The existing WhatsApp and paper-based workflow was analysed to identify operational pain points. The analysis focused on how animal cases, treatment updates, donations, adoption posts, and medicine stock details are currently handled.

3.1.4 Document and Record Analysis

Existing record formats, informal message patterns, adoption posts, donation tracking methods, and medicine stock practices were reviewed to identify data fields and workflow requirements for the proposed system.

3.2 Existing System Analysis

Animal Ark currently manages many activities through informal and manual methods such as WhatsApp messages, paper notes, spreadsheets, social media posts, and verbal communication. Although these methods are familiar to volunteers, they create several problems when the number of animal cases, volunteers, donations, and adoption requests increases.

3.2.1 Current Workflow Summary

1. A rescued or injured animal is reported through a message, phone call, or direct contact.
2. Volunteers coordinate rescue and treatment through WhatsApp or verbal communication.
3. Treatment updates are shared informally using messages, photos, or handwritten notes.
4. Handover details between volunteers are often communicated manually.
5. Emergency situations are handled through urgent calls or WhatsApp messages.
6. Medicine stock is checked manually or by asking responsible members.
7. Donations are promoted through posts or messages and recorded manually.
8. Adoption posts are shared through social media or informal communication.
9. Volunteer contributions are not always recorded in a structured manner.

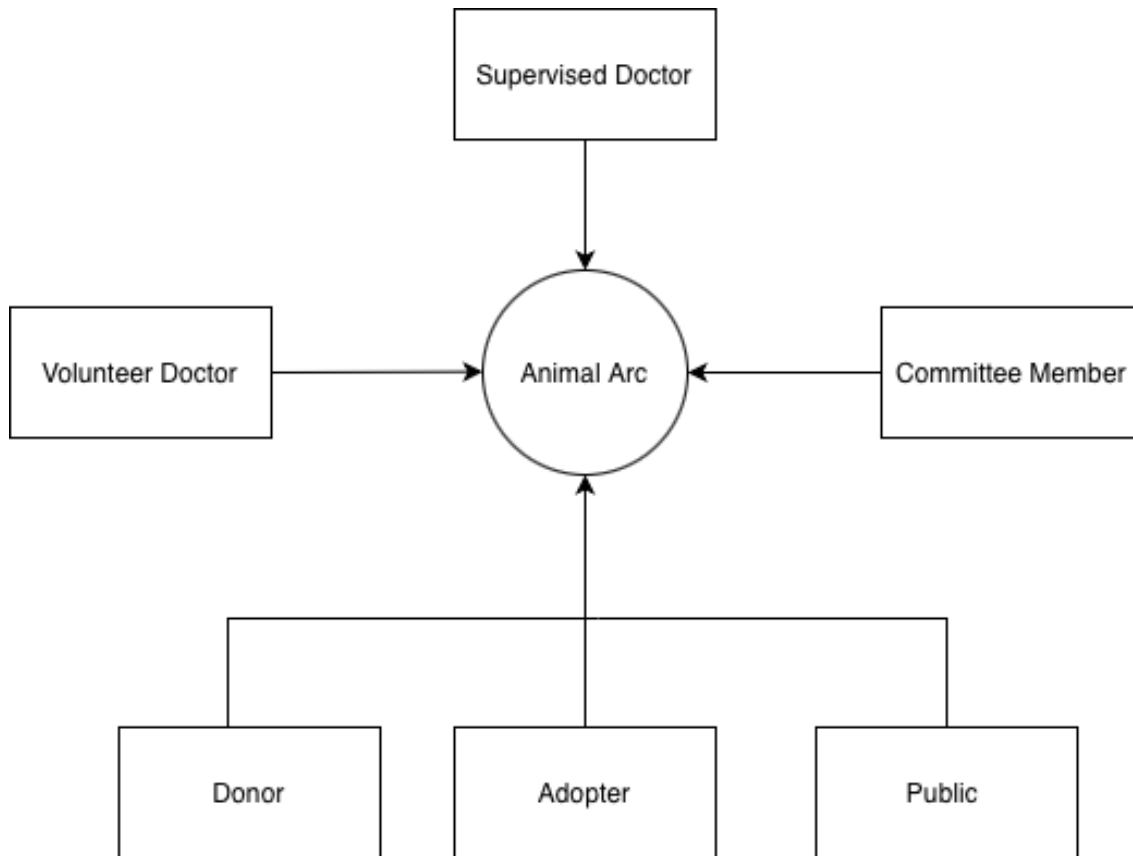
3.2.2 Key Pain Points Identified

- Missing or Scattered Treatment Records: Treatment details may be spread across WhatsApp chats, paper notes, or individual memory.
- Weak Handover Communication: Important instructions may be missed when volunteers change shifts.
- Emergency Blind Spots: Urgent cases may not reach all responsible volunteers and supervisors quickly.
- Medicine Stock Uncertainty: Volunteers may not know whether required medicines are available.
- Manual Donation Tracking: Donation records may be difficult to verify, summarize, or link to campaigns.
- Unstructured Adoption Flow: Adoption interest, screening, and approval may be handled informally.
- Lack of Searchable Animal History: Previous treatment details and case outcomes may be difficult to retrieve.
- Limited Volunteer Recognition: Volunteer work and contribution statistics are not consistently recorded.
- Public Engagement Gaps: Donors, adopters, and supporters do not have one centralized portal to interact with Animal Ark.

3.2.3 Proposed Improvement

Animal Ark will address these problems by providing one centralized system for animal profiles, treatment timelines, handovers, emergency alerts, medicine stock, donations, blood donors, adoptions, volunteer statistics, and public engagement.

3.3 Context Diagram



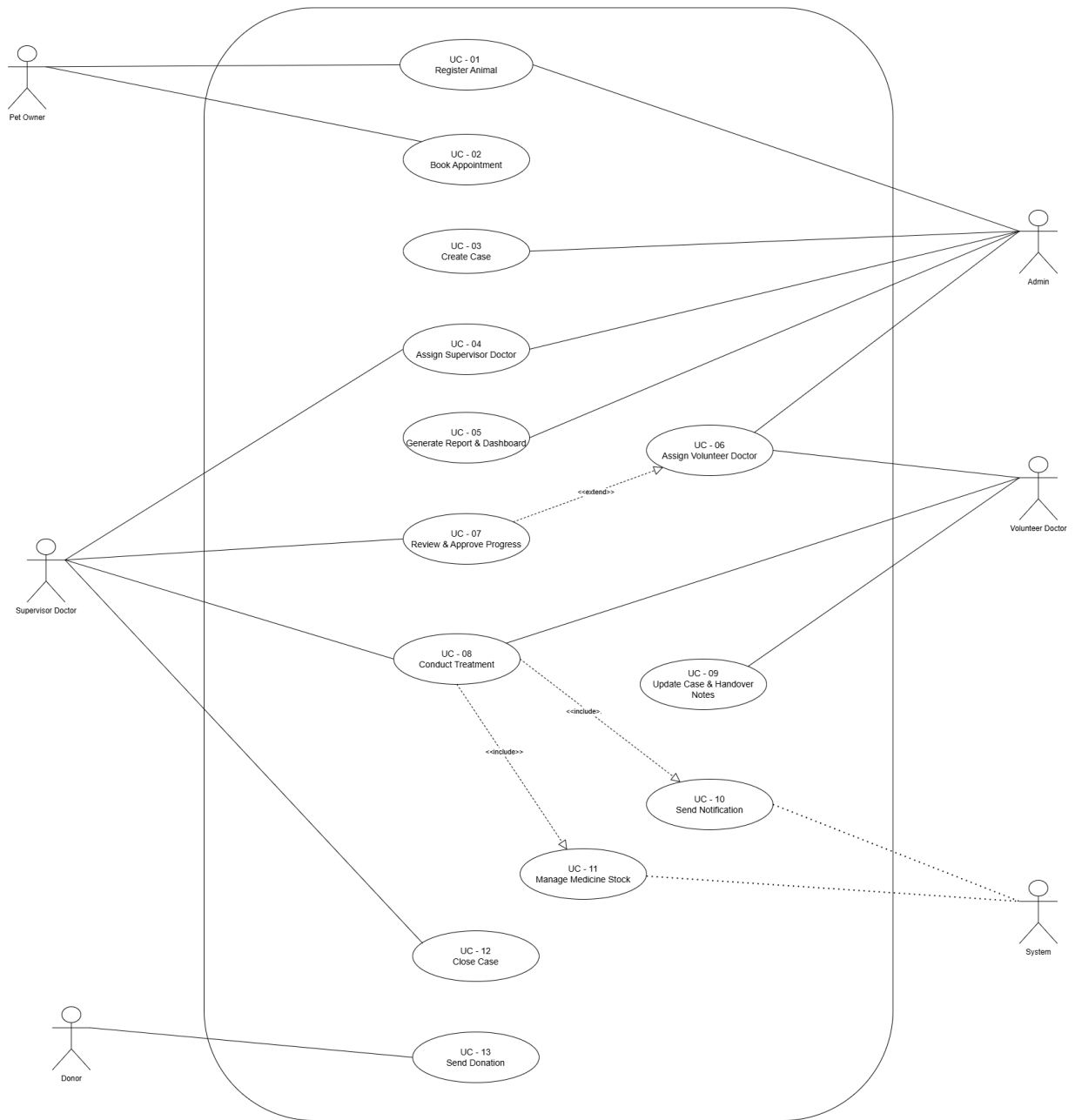
3.4 Use Case Diagram

The use case diagram should show the Animal Arc system boundary and the major actors that interact with the system.

The main actors are:

- Volunteer Veterinary Student
- Supervisor Doctor
- Committee Admin
- Donor
- Adopter
- Public Visitor

Veterinary Society Management System



3.5 Use Case Descriptions

This section provides detailed use case descriptions for all 13 use cases identified in the Animal Arc system. Each description includes the use case ID, actor(s), preconditions, main flow, alternative flows, postconditions, and business rules.

UC-01: Animal Intake and Profile Creation

Field	Details
Use Case ID	UC-01
Use Case Name	Animal Intake and Profile Creation
Actor(s)	Primary: Intake Staff / Volunteer Veterinary Student Secondary: Committee Admin
Description	Allows an authorized staff member or volunteer to register a newly rescued or brought-in street animal into the system by capturing all intake details.
Preconditions	The user is logged into the system with Intake Staff or Volunteer role. The animal has been physically brought to the care Centre or rescue location is known. The system is accessible via browser or mobile device.
Postconditions	A new animal profile is created in the system with a unique Animal ID. An intake record is saved with the bringer's details, found location, and initial condition. The animal status is set to 'Active – Awaiting Medical Case'. The animal profile is visible to authorized internal users.
Main Flow	1. User selects 'Register New Animal' from the dashboard. 2. System displays the Animal Intake Form. 3. User enters animal species, estimated age, sex, color, identifying marks, and initial condition.

	4. User enters the rescue/found location and the name and contact details of the person who brought the animal. 5. User selects the urgency level (Normal, Urgent, Critical). 6. User uploads one or more animal photos. 7. User enters any initial notes about the animal's condition. 8. User submits the intake form. 9. System validates all required fields. 10. System generates a unique Animal ID and saves the profile. 11. System displays a success message with the new Animal ID.
Alternative Flows	A1 – Missing Required Fields: At step 9, if required fields (species, condition, urgency) are empty, system highlights the missing fields and returns form to user. A2 – Photo Upload Failure: At step 6, if the uploaded photo exceeds the size limit or is an unsupported format, system displays an error and allows the user to re-upload or skip photos. A3 – Network Interruption: If the form submission fails due to a network issue, system retains the entered data and prompts the user to retry.
Business Rules	Species, initial condition, and urgency level are mandatory fields. Each animal is assigned a unique system-generated Animal ID. Photos must not exceed 2 MB per image. Intake location must be recorded as either a map coordinate or a text description.

UC-02: Treatment Timeline Management

Field	Details
Use Case ID	UC-02
Use Case Name	Treatment Timeline Management
Actor(s)	Primary: Volunteer Veterinary Student (Volunteer Doctor) Secondary: Supervisor Doctor
Description	Allows the assigned volunteer doctor to add, update, and view treatment timeline entries for an active animal medical case. Each entry records the treatment stage, medicines used, observations, and next actions.
Preconditions	The user is logged in with Volunteer Doctor or Supervisor Doctor role. An animal profile and medical case already exist in the system. The volunteer doctor has been assigned to the case by the Supervisor Doctor. The case status is 'Open' or 'In Progress'.
Postconditions	A new treatment entry is saved to the case's treatment timeline. Medicine stock is reduced if medicine usage is recorded. The case's last updated timestamp is refreshed. The Supervisor Doctor can view the new entry in the case overview.
Main Flow	1. User navigates to their assigned case list. 2. User selects the animal case to update. 3. System displays the case profile and existing treatment timeline. 4. User selects 'Add Treatment Entry'. 5. System displays the treatment entry form.

	6. User selects the treatment stage (e.g., Initial Checkup, Diagnosis Added, Treatment Started, Follow-up Treatment, Recovery Review). 7. User enters diagnosis notes, treatment observations, and next action. 8. User records any medicines used and the quantity consumed. 9. User uploads supporting photos or documents if applicable. 10. User sets the entry status (Pending / Ongoing / Completed). 11. User submits the treatment entry. 12. System validates required fields. 13. System saves the timeline entry linked to the case and doctor. 14. System reduces medicine stock quantities for recorded medicine usage. 15. System displays updated treatment timeline to the user.
Alternative Flows	A1 – Missing Required Fields: At step 12, if the treatment stage or notes are empty, system highlights missing fields and returns the form. A2 – Insufficient Medicine Stock: At step 14, if the recorded medicine quantity exceeds available stock, system displays a stock warning and asks the user to confirm or adjust the quantity. A3 – Case Already Closed: At step 4, if the case status is 'Closed', system prevents adding new entries and notifies the user that the case is no longer active.
Business Rules	Treatment stage and notes are mandatory for each entry. Medicine usage entries must reference an existing medicine in the stock registry. Only the assigned volunteer doctor or supervisor doctor can add entries to a case. Each entry is timestamped and linked to the doctor who submitted it.

	Timeline entries cannot be deleted; they can only be marked as corrected with a note.
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UC-03: Handover Notes Management

Field	Details
Use Case ID	UC-03
Use Case Name	Handover Notes Management
Actor(s)	Primary: Volunteer Veterinary Student Secondary: Supervisor Doctor
Description	Allows a volunteer to add handover notes to an active animal case so that the incoming volunteer or supervisor is informed of pending tasks, medicine schedules, feeding instructions, warnings, and important observations before taking over.
Preconditions	The user is logged in with Volunteer or Supervisor Doctor role. The animal case is active and in progress. The current shift volunteer is responsible for the case.
Postconditions	A handover note is saved and linked to the animal case. The handover note is visible to the incoming volunteer or supervisor when they next view the case. The handover note is timestamped and attributed to the submitting volunteer.
Main Flow	1. User opens the active animal case. 2. User selects 'Add Handover Note'. 3. System displays the handover note form. 4. User enters a summary of the current status. 5. User records pending tasks and medicine schedule reminders. 6. User adds feeding instructions and any warnings or critical observations.

	7. User submits the handover note. 8. System validates that note content is not empty. 9. System saves the handover note with timestamp and author information. 10. System displays the updated handover notes list for the case.
Alternative Flows	A1 – Empty Note Submission: At step 8, if the note content is empty, system returns the form with a validation message. A2 – Case Closed: If the case has been closed, system prevents adding new handover notes and informs the user.
Business Rules	Handover note content must not be empty. Notes are read-only once submitted; corrections require a new note. Each note must display the author's name and submission timestamp. Notes are visible in reverse chronological order (most recent first).

UC-04: Supervisor Oversight and Review

Field	Details
Use Case ID	UC-04
Use Case Name	Supervisor Oversight and Review
Actor(s)	Primary: Supervisor Doctor Secondary: Volunteer Veterinary Student
Description	Allows a Supervisor Doctor to review active animal cases, examine treatment timelines, add supervisor comments or instructions, assign or reassign volunteer doctors, and approve or reject case updates and recovery decisions.
Preconditions	The user is logged in with Supervisor Doctor role. At least one active animal case exists in the system. Treatment timeline entries have been submitted by the assigned volunteer doctor.
Postconditions	Supervisor comments are saved and linked to the case. Doctor assignment or reassignment is recorded. If recovery is confirmed, case status is updated to 'Recovered – Awaiting Adoption Decision'. Volunteer doctors are notified of supervisor feedback or instructions.
Main Flow	1. Supervisor logs in and opens the Supervisor Dashboard. 2. System displays all active cases with current status and urgency. 3. Supervisor selects a case to review. 4. System displays the full case profile and treatment timeline. 5. Supervisor reviews treatment entries, handover notes, and current animal status. 6. Supervisor adds a supervisor comment or medical instruction. 7. Supervisor may assign or reassign a volunteer doctor to the case.

	8. Supervisor submits the review. 9. System saves the comment, updates the assignment if changed, and notifies the relevant volunteer doctor. 10. If the Supervisor confirms recovery, system updates the case status to 'Recovered'.
Alternative Flows	A1 – No Active Cases: At step 2, if no active cases exist, system displays an empty state message. A2 – Recovery Rejection: At step 10, if the Supervisor determines the animal has not recovered, the case remains 'In Progress' and the supervisor adds instructions for further treatment. A3 – No Volunteer Doctor Available: At step 7, if no eligible volunteer doctors are registered, system displays a message and the case remains unassigned.
Business Rules	Only users with Supervisor Doctor role can approve recovery and assign doctors. Supervisor comments are timestamped and cannot be deleted. A case can have only one assigned volunteer doctor at a time. Recovery confirmation is a required step before an adoption notice can be published.

UC-05: Case Closure

Field	Details
Use Case ID	UC-05
Use Case Name	Case Closure
Actor(s)	Primary: Supervisor Doctor Secondary: Committee Admin
Description	Allows an authorised user to formally close an animal medical case by selecting the closure reason and recording final notes. Closure reasons include recovered, adopted, released into the wild, transferred to another facility, or deceased.
Preconditions	The user is logged in with Supervisor Doctor or Committee Admin role. The animal case is in an active or recovered state. All required treatment timeline entries are complete. If closure reason is 'Adopted', an approved adoption application must exist.
Postconditions	The case status is set to 'Closed'. The closure reason, final notes, and closure date are recorded. The animal profile is updated to reflect the final outcome. No further treatment entries or handover notes can be added to the closed case. The case is archived but remains searchable for historical reference.
Main Flow	1. User navigates to the animal case. 2. User selects 'Close Case'. 3. System displays the case closure form. 4. User selects the closure reason (Recovered, Adopted, Released, Transferred, Deceased). 5. User enters final notes or observations.

	6. User confirms the closure action. 7. System validates that a closure reason is selected. 8. System updates the case status to 'Closed' and records the closure date. 9. System updates the animal's current status on the animal profile. 10. System displays a confirmation message.
Alternative Flows	A1 – Missing Closure Reason: At step 7, if no closure reason is selected, system prevents closure and returns the form. A2 – Pending Adoption Application: If closure reason is 'Adopted' but no approved adoption application exists, system warns the user and requires adoption confirmation before proceeding. A3 – Critical Open Timeline Items: If the treatment timeline has entries marked as 'Pending', system warns the user and asks for confirmation to proceed with closure despite pending items.
Business Rules	Closure reason is a mandatory field. Closed cases cannot have new treatment entries added. Closed cases remain in the system as read-only historical records. Case closure triggers archival of all related handover notes and timeline entries.

UC-06: Emergency Alert Management

Field	Details
Use Case ID	UC-06
Use Case Name	Emergency Alert Management
Actor(s)	Primary: Volunteer Veterinary Student / Committee Admin Secondary: Supervisor Doctor
Description	Allows an authorised user to create and broadcast an emergency alert for urgent rescue or treatment situations. Emergency alerts notify relevant volunteers and supervisors who can acknowledge and update the alert status.
Preconditions	The user is logged in with Volunteer or Committee Admin role. An urgent situation requiring immediate response has occurred. The system notification service is operational.
Postconditions	An emergency alert is created and visible on the dashboard of all relevant users. Relevant volunteers and supervisor doctors receive a notification. The alert status is set to 'Open'. Alert responses and status changes are tracked.
Main Flow	1. User selects 'Create Emergency Alert' from the dashboard. 2. System displays the emergency alert form. 3. User selects the emergency type (Rescue Needed, Critical Treatment, Blood Donation Needed, Medicine Shortage, Other). 4. User enters a description of the emergency, location, and required support. 5. User links the alert to an animal case if applicable. 6. User submits the alert. 7. System validates required fields.

	8. System creates the alert with status 'Open' and broadcasts a notification to relevant users. 9. Notified users can view the alert on their dashboard. 10. A responding user acknowledges the alert and updates the status to 'Assigned' or 'In Progress'. 11. Once the emergency is resolved, an authorised user marks the alert as 'Resolved'.
Alternative Flows	A1 – Missing Emergency Type or Description: At step 7, if required fields are empty, system returns the form with validation errors. A2 – No Users to Notify: If no eligible users are registered to receive the alert type, system still creates the alert and displays it on the dashboard. A3 – Alert Cancelled: If the situation is resolved before a response, the creator can cancel the alert and record a cancellation note.
Business Rules	Emergency type and description are mandatory. Only the alert creator or a Committee Admin can cancel or close an emergency alert. Alert status must follow the sequence: Open → Assigned → In Progress → Resolved / Cancelled. Emergency alerts remain in the system as a log even after resolution.

UC-07: Medicine Stock Management

Field	Details
Use Case ID	UC-07
Use Case Name	Medicine Stock Management
Actor(s)	Primary: Committee Admin Secondary: Volunteer Doctor / Supervisor Doctor
Description	Allows the Committee Admin to add, update, and manage the medicine and medical supply inventory. Volunteers and supervisors can view stock levels when recording medicine usage in treatment entries. The system triggers low-stock alerts when quantities fall below the defined minimum threshold.
Preconditions	The user is logged in with Committee Admin role (for add/update/delete). At least the medicine name, quantity, and unit are available to enter. For low-stock alert: a minimum stock threshold has been defined for the medicine.
Postconditions	The medicine record is created or updated in the inventory. Stock quantities are adjusted when treatment entries record medicine usage. A low-stock notification is sent when quantity falls below the defined threshold. All stock changes are logged with the user who made the change and the timestamp.
Main Flow	1. Admin navigates to the Medicine Inventory page. 2. System displays the current medicine inventory list. 3. Admin selects 'Add New Medicine'. 4. System displays the medicine entry form. 5. Admin enters medicine name, category, quantity, unit, expiry date, storage notes, and minimum stock threshold. 6. Admin submits the form. 7. System validates required fields.

	8. System saves the medicine record and displays the updated inventory. 9. When a treatment entry records medicine usage, system automatically reduces the stock quantity. 10. If stock falls below the minimum threshold, system triggers a low-stock alert notification to the Committee Admin.
Alternative Flows	A1 – Duplicate Medicine Name: At step 7, if a medicine with the same name already exists, system warns the admin and prompts to update the existing record instead. A2 – Expired Medicine: System highlights medicines with an expiry date that has passed and flags them for review or removal. A3 – Stock Update from Treatment Entry: If the treatment entry records a quantity greater than available stock, system displays a warning and asks the doctor to confirm or adjust the amount.
Business Rules	Medicine name, quantity, and unit are mandatory fields. Stock can only be reduced by recording medicine usage in a treatment entry. Only Committee Admin can add, edit, or remove medicine records. The system must log every stock change with user, action, and timestamp. Medicines with quantity at or below the minimum threshold trigger a low-stock alert.

UC-08: Donation Campaign and Donation Record Management

Field	Details
Use Case ID	UC-08
Use Case Name	Donation Campaign and Donation Record Management
Actor(s)	Primary: Committee Admin Secondary: Donor (Public User)
Description	Allows the Committee Admin to create and manage donation campaigns for specific animal cases, medicine needs, or general welfare support. Donors can view campaigns through the public portal and submit donation interest. Admins record received donations and link them to campaigns.
Preconditions	The Committee Admin is logged in. A need for donations has been identified (for a specific animal case or general supply). For donation submission: the public portal is accessible to the donor.
Postconditions	A donation campaign is created and visible on the public portal. Received donations are recorded and linked to the campaign. Campaign progress (amount raised vs target) is updated. Donation records are stored with donor details and proof of donation.
Main Flow	1. Admin selects 'Create Donation Campaign' from the dashboard. 2. System displays the campaign creation form. 3. Admin enters campaign title, description, required items or monetary target, target date, and links to an animal case if applicable. 4. Admin sets the campaign status to 'Active'. 5. Admin submits the campaign. 6. System validates required fields and saves the campaign.

	7. Campaign becomes visible on the public portal. 8. A donor views the campaign on the public portal and submits donation interest or details. 9. Admin records the received donation with donor name, contact, donation type, amount or item, date, and proof of donation. 10. System updates the campaign's received donations summary. 11. Admin closes the campaign when the target is met or the deadline passes.
Alternative Flows	A1 – Missing Campaign Fields: At step 6, if the campaign title or description is empty, system returns the form. A2 – Donation Submitted Without Campaign: If a donor contacts Animal Arc outside the system, Admin can record the donation as a general donation not linked to a specific campaign. A3 – Campaign Target Already Met: System notifies the admin when the donation target is reached and the admin can update the campaign status to 'Target Met'.
Business Rules	Campaign title and description are mandatory. Only Committee Admin can create, edit, or close campaigns. Each donation record must include the donor name, donation type, and date at minimum. Proof of donation image upload is recommended but optional. Campaign status values: Active, Target Met, Closed.

UC-09: Blood Donor Registration and Search

Field	Details
Use Case ID	UC-09
Use Case Name	Blood Donor Registration and Search
Actor(s)	Primary: Public Visitor / Animal Owner (Registration) Secondary: Volunteer Doctor / Supervisor Doctor (Search)
Description	Allows animal owners or authorised staff to register an animal as a potential blood donor. Allows volunteer and supervisor doctors to search the blood donor registry during emergency transfusion situations by filtering by species, location, and availability.
Preconditions	For registration: The registrant has the animal's details and the owner's contact information. For search: The user is logged in with Volunteer or Supervisor Doctor role. A blood transfusion emergency exists that requires a donor search.
Postconditions	A new blood donor record is saved in the registry. The donor record is searchable by authorised internal users. Search results are returned filtered by the specified criteria. Contact details of matching donors are available to the searching doctor.
Main Flow – Registration	1. Animal owner or authorised staff accesses the blood donor registration form. 2. Registrant enters animal species, breed, estimated age, weight, and blood type if known. 3. Registrant enters owner name, contact phone, and location. 4. Registrant sets availability status (Available / Unavailable). 5. Registrant submits the form. 6. System validates required fields and saves the blood donor record.

	7. System confirms successful registration.
Main Flow – Search	1. Doctor navigates to the Blood Donor Search page. 2. Doctor enters search criteria: species, location, blood type, and availability. 3. System retrieves and displays matching donor records. 4. Doctor views donor contact details and availability for the selected donor. 5. Doctor contacts the owner externally using the displayed contact details.
Alternative Flows	A1 – No Matching Donors Found: At step 3 of Search, if no donors match the criteria, system displays a 'No donors found' message and suggests broadening the search. A2 – Incomplete Registration: At step 6 of Registration, if required fields are missing, system returns the form with highlighted errors.
Business Rules	Animal species, owner contact, and location are mandatory for registration. Blood donor records can be updated or deactivated by the owner or an admin. Only logged-in volunteers and supervisors can view full donor contact details. Public users can access the registration form but cannot view the registry.

UC-10: Adoption Flow Management

Field	Details
Use Case ID	UC-10
Use Case Name	Adoption Flow Management
Actor(s)	Primary: Committee Admin / Supervisor Doctor (Listing) Secondary: Adopter / Public User (Application)
Description	Covers the full adoption process: an authorised user creates an adoption listing for a recovered animal, the listing is published on the public portal, public users submit adoption applications, and the Committee Admin reviews and approves or rejects applications.
Preconditions	For listing creation: The animal's case status must be 'Recovered' and recovery must be confirmed by the Supervisor Doctor. For application submission: The adoption listing is published and visible on the public portal. For application review: The Committee Admin is logged in.
Postconditions	The adoption listing is saved and visible on the public portal. An adoption application is submitted and linked to the listing. The application status is updated (Approved / Rejected / Under Review). If approved, the case is eligible for closure with reason 'Adopted'. The adopter receives a status notification.
Main Flow	1. Supervisor Doctor confirms recovery and marks the animal as 'Ready for Adoption'. 2. Admin navigates to the Adoption Listings page and selects 'Create Adoption Notice'. 3. System displays the adoption listing form.

	4. Admin enters animal description, health status, temperament, adoption requirements, and uploads photos. 5. Admin sets the listing status to 'Published'. 6. System saves the listing and makes it visible on the public portal. 7. A public user (Adopter) views the listing and selects 'Apply to Adopt'. 8. System displays the adoption application form. 9. Adopter enters their name, contact details, home address, living environment description, and reason for adopting. 10. Adopter submits the application. 11. System saves the application with status 'Under Review'. 12. Admin reviews the application and approves or rejects it. 13. System updates the application status and notifies the adopter.
Alternative Flows	A1 – Animal Not Marked as Recovered: If the Supervisor has not confirmed recovery, the adoption listing creation option is disabled. A2 – Application Incomplete: At step 10, if required applicant fields are missing, system returns the form. A3 – Multiple Applications for One Animal: System allows multiple applications per listing. Only one application can be approved; approval of one automatically closes others as 'Unsuccessful'. A4 – Listing Withdrawn: Admin can unpublish or withdraw a listing if the animal is no longer available for adoption.

Business Rules	An adoption listing can only be created after the Supervisor Doctor confirms recovery. At least one photo must be included in the adoption listing. Application approval requires Committee Admin authorisation. Approving one application automatically sets all other applications for the same animal to 'Unsuccessful'. Application status values: Under Review, Approved, Rejected, Unsuccessful.
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UC-11: Volunteer Statistics Tracking

Field	Details
Use Case ID	UC-11
Use Case Name	Volunteer Statistics Tracking
Actor(s)	Primary: Volunteer Veterinary Student Secondary: Committee Admin / Supervisor Doctor
Description	The system automatically tracks and displays volunteer contribution statistics based on system activity. Volunteers can view their own statistics, while supervisors and admins can view statistics for all volunteers.
Preconditions	The volunteer is logged in. The volunteer has performed actions in the system (treatment entries, handover notes, emergency responses, etc.). The statistics module is enabled.
Postconditions	Volunteer statistics are calculated and displayed accurately. Statistics reflect all system activities linked to the volunteer. Supervisors and admins can access a summary view across all volunteers.
Main Flow	1. Volunteer navigates to the 'My Statistics' page. 2. System retrieves all activity records linked to the logged-in volunteer. 3. System calculates and displays statistics: number of cases handled, treatment entries added, handover notes submitted, emergency alerts responded to, and adoption support activities. 4. Volunteer views a summary and breakdown of their contributions. 5. Committee Admin or Supervisor Doctor can navigate to the Volunteer Statistics overview page to compare contributions across all volunteers.
Alternative Flows	A1 – No Activity Yet:

	If the volunteer has not performed any actions, system displays zero counts with a message to begin contributing. A2 – Admin View: Admin can filter volunteer statistics by date range, case type, or specific volunteer.
Business Rules	Statistics are calculated automatically from system activity logs; manual editing is not permitted. Volunteers can only see their own statistics. Supervisors and admins can see all volunteers' statistics. Statistics are updated in real time as activities are submitted.

UC-12: Public Portal Access

Field	Details
Use Case ID	UC-12
Use Case Name	Public Portal Access
Actor(s)	Primary: Public Visitor Secondary: Donor / Adopter
Description	Allows any public user to access the Animal Arc public portal without requiring login. The portal provides access to published adoption listings, active donation campaigns, blood donor registration, awareness content, and Animal Ark contact information. Public users can submit adoption applications, donation interest, blood donor registrations, and general inquiries.
Preconditions	The public portal is accessible via a web browser. At least one adoption listing or donation campaign is published. No login is required for public portal browsing.
Postconditions	Public user views the requested content. Submitted inquiries, donation interests, blood donor registrations, or adoption applications are saved. Internal records are not exposed to public users. The admin receives a notification for new public submissions where applicable.
Main Flow	1. Public user opens the Animal Arc public portal URL. 2. System displays the public home page with adoption listings, donation campaigns, and awareness content. 3. User browses available adoption listings. 4. User selects an animal to view the full adoption listing details. 5. User selects 'Apply to Adopt' and submits an adoption application (see UC-10).

	6. Alternatively, user views a donation campaign and submits a donation inquiry. 7. Alternatively, user accesses the blood donor registration form and submits a registration (see UC-09). 8. Alternatively, user fills in the general contact or inquiry form and submits. 9. System confirms submission with a thank-you message.
Alternative Flows	A1 – No Listings Available: If no adoption listings or campaigns are published, system displays a notice that no items are currently available. A2 – Inquiry Form Submission Failure: If submission fails due to a network issue, system retains form data and prompts the user to retry. A3 – Attempt to Access Internal Pages: If a public user attempts to access internal pages (e.g., case management), system redirects to the login page.
Business Rules	The public portal must not expose internal treatment records, volunteer details, or donor contact information. Public users cannot create, edit, or delete any internal records. Adoption applications submitted through the portal are saved with 'Under Review' status. All public submissions are visible to Committee Admin for follow-up.

UC-13: User Authentication and Role Management

Field	Details
Use Case ID	UC-13
Use Case Name	User Authentication and Role Management
Actor(s)	Primary: Committee Admin Secondary: All Internal Users (Volunteer, Supervisor Doctor)
Description	Covers secure login, logout, and role-based access control for all internal users. The Committee Admin can register new internal users, assign roles, activate or deactivate accounts, and update role assignments. All protected system features require authentication.
Preconditions	For login: A registered user account exists with a valid email and password. For user management: The Committee Admin is logged in. The authentication service (JWT) is operational.
Postconditions	Authenticated user is granted access to features permitted for their role. New user accounts are created and role-assigned by the admin. Deactivated accounts cannot log in. All login attempts are logged for audit purposes.
Main Flow – Login	1. User navigates to the login page. 2. User enters their registered email and password. 3. User submits the login form. 4. System validates credentials against the stored user records. 5. System generates a JWT token and establishes a session. 6. System redirects the user to the dashboard appropriate for their role.

Main Flow – User Management	1. Admin navigates to the User Management page. 2. System displays the list of all registered users with their roles and statuses. 3. Admin selects 'Add New User'. 4. Admin enters the user's name, email, phone number, and assigns a role. 5. System creates the account and sends login credentials to the new user. 6. Admin can update an existing user's role or deactivate their account. 7. System saves the changes and reflects the updated role immediately.
Alternative Flows	A1 – Incorrect Credentials: At step 4 of Login, if credentials do not match, system displays an error message. After 5 failed attempts, the account is temporarily locked. A2 – Deactivated Account: At step 4, if the user's account is deactivated, system displays an 'Account Inactive' message and instructs the user to contact the admin. A3 – Duplicate Email: At step 5 of User Management, if the email already exists in the system, system prevents creation and notifies the admin.
Business Rules	Passwords must be stored as hashed values; plaintext storage is not permitted. Each user must have exactly one role: Admin, Supervisor Doctor, Volunteer Doctor, Intake Staff, Donor, or Adoption Applicant. Only Committee Admin can create, update, or deactivate user accounts. JWT tokens expire after a defined period; users must re-authenticate after expiry. Role changes take effect immediately without requiring the affected user to log out and back in.

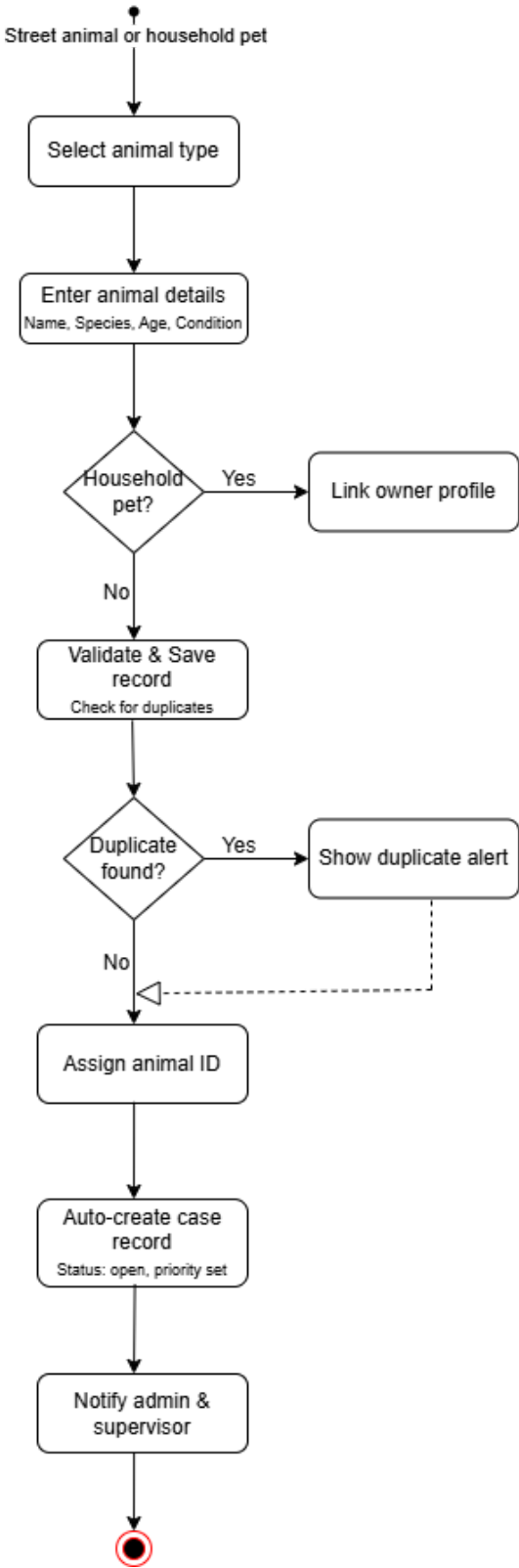
3.6 Activity Diagrams

Activity diagrams should be created for the major workflows of the Animal Arc system. The diagrams should include main flows, alternative flows, validation steps, and decision points.

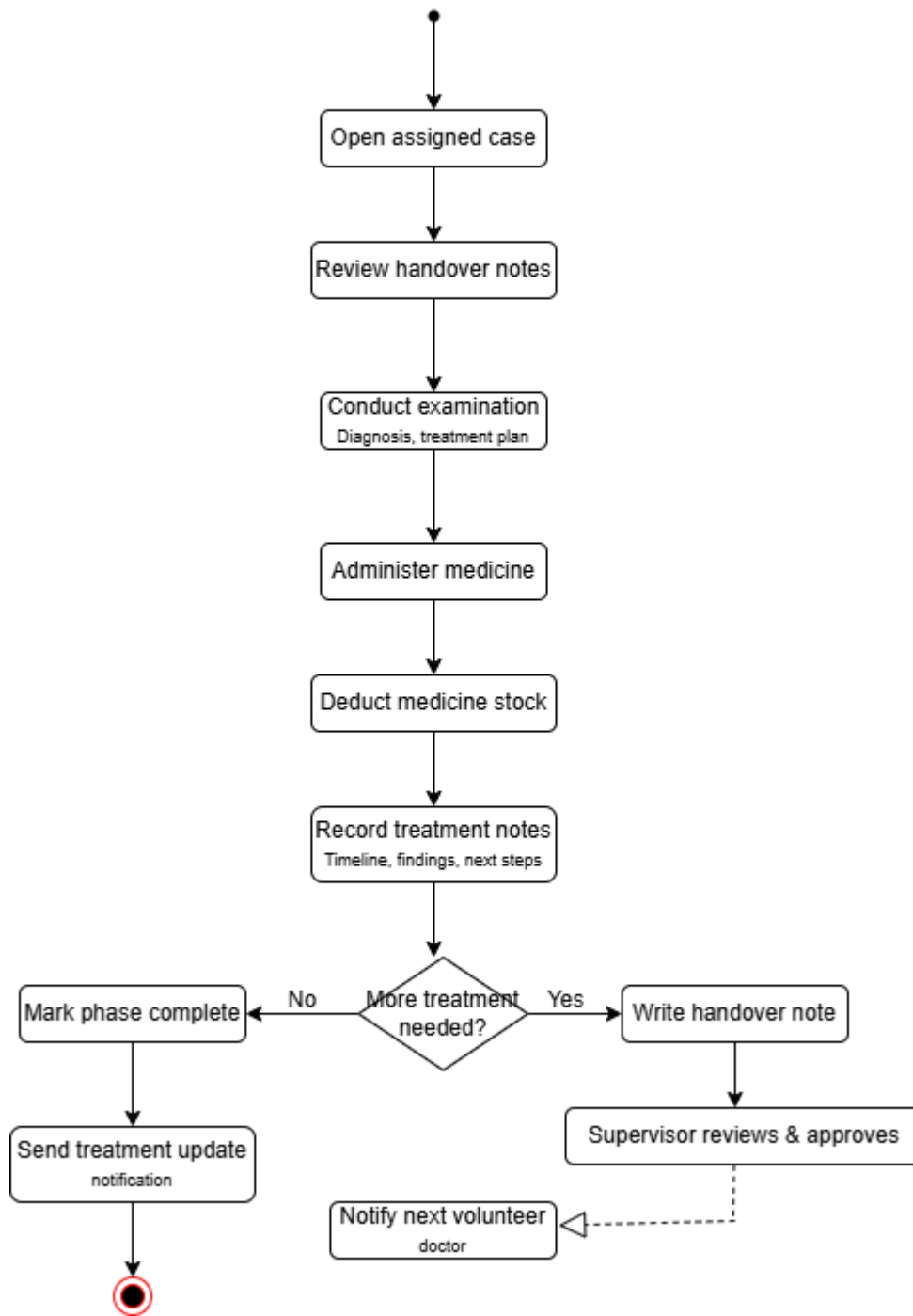
Recommended activity diagrams:

- animal-intake-flow.drawio
- treatment-timeline-flow.drawio
- handover-notes-flow.drawio
- emergency-alert-flow.drawio
- medicine-stock-flow.drawio
- donation-campaign-flow.drawio
- blood-donor-registration-flow.drawio
- adoption-process-flow.drawio
- case-closure-flow.drawio
- public-portal-flow.drawio

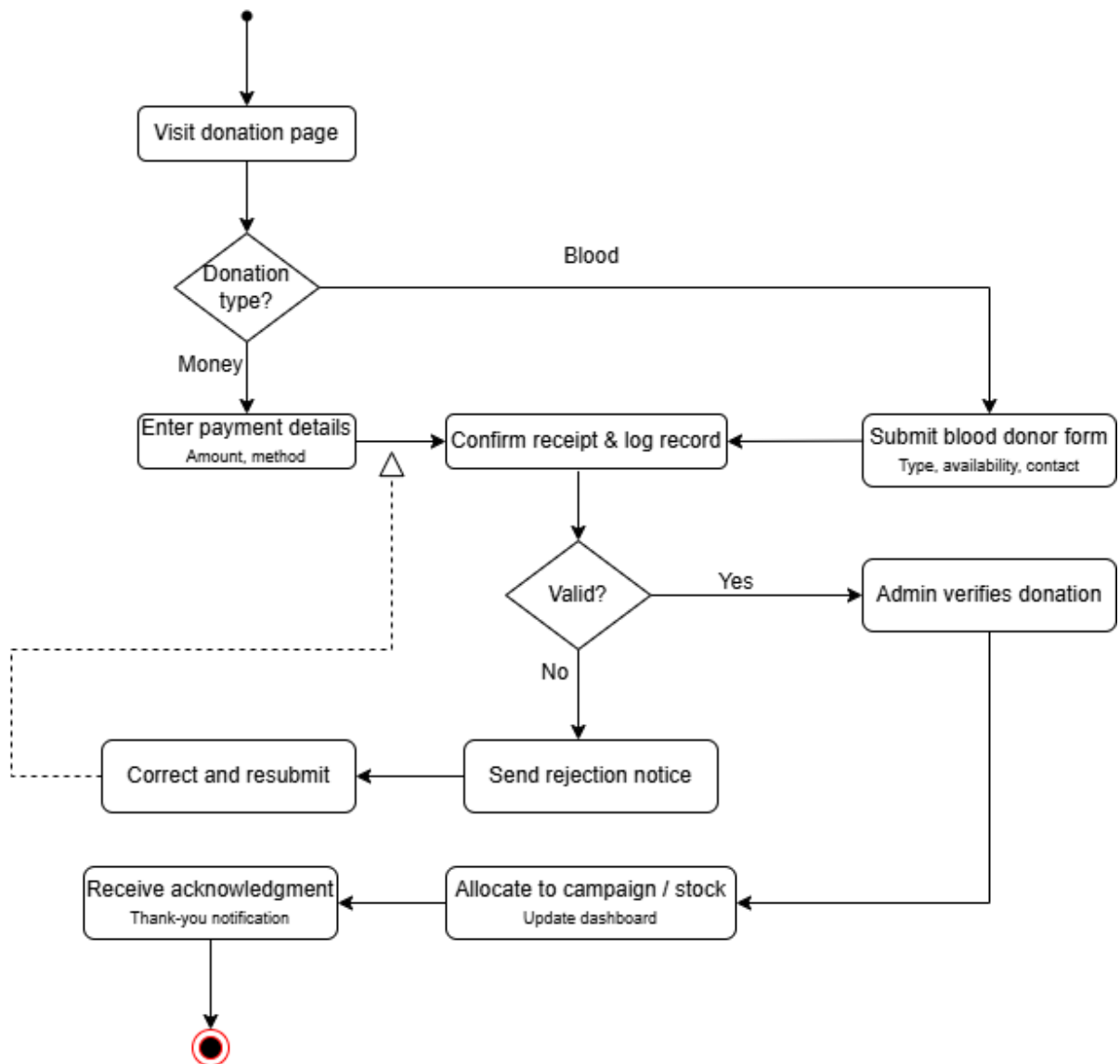
Animal Registration Flow



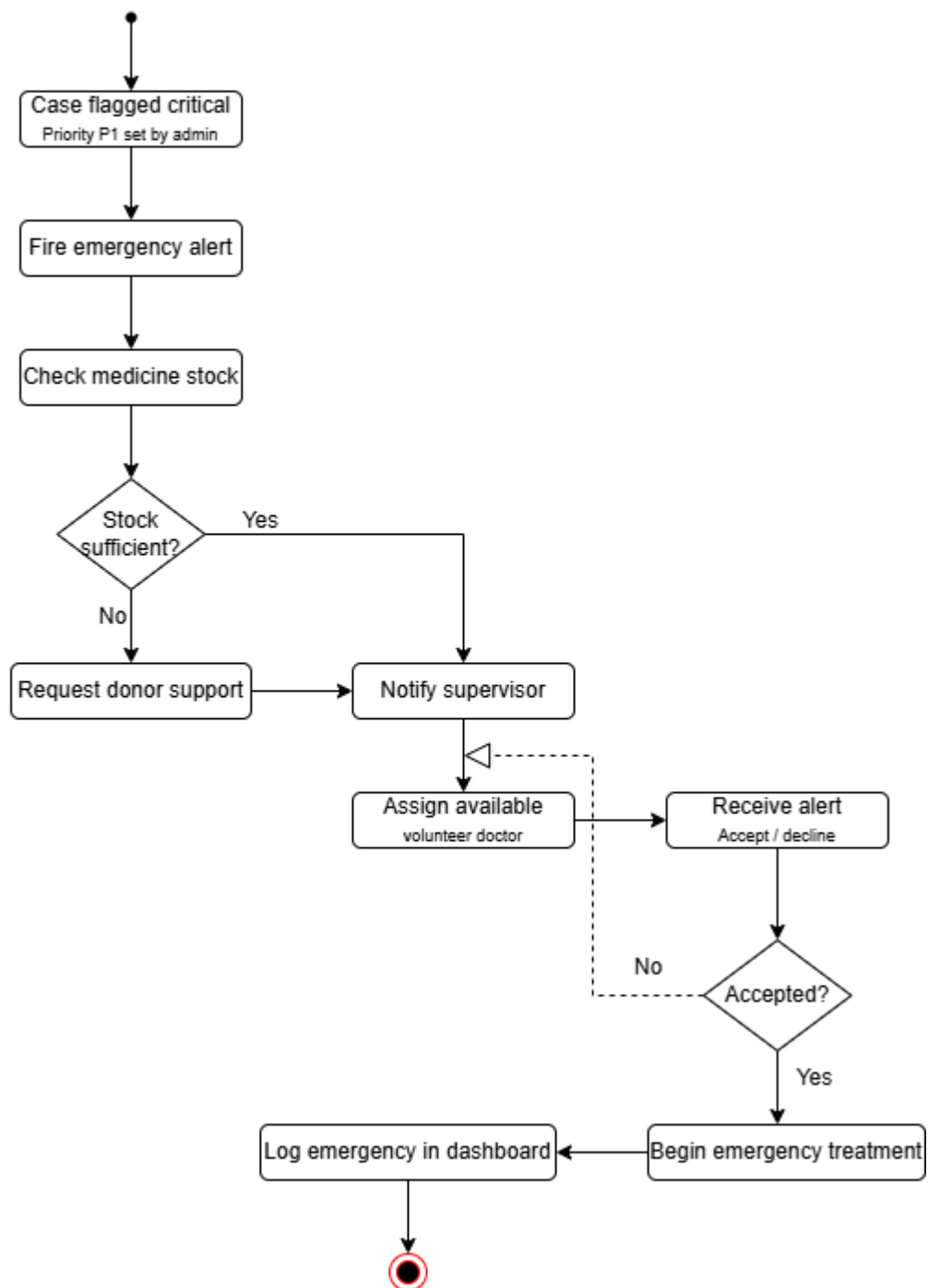
Treatment Recording Flow



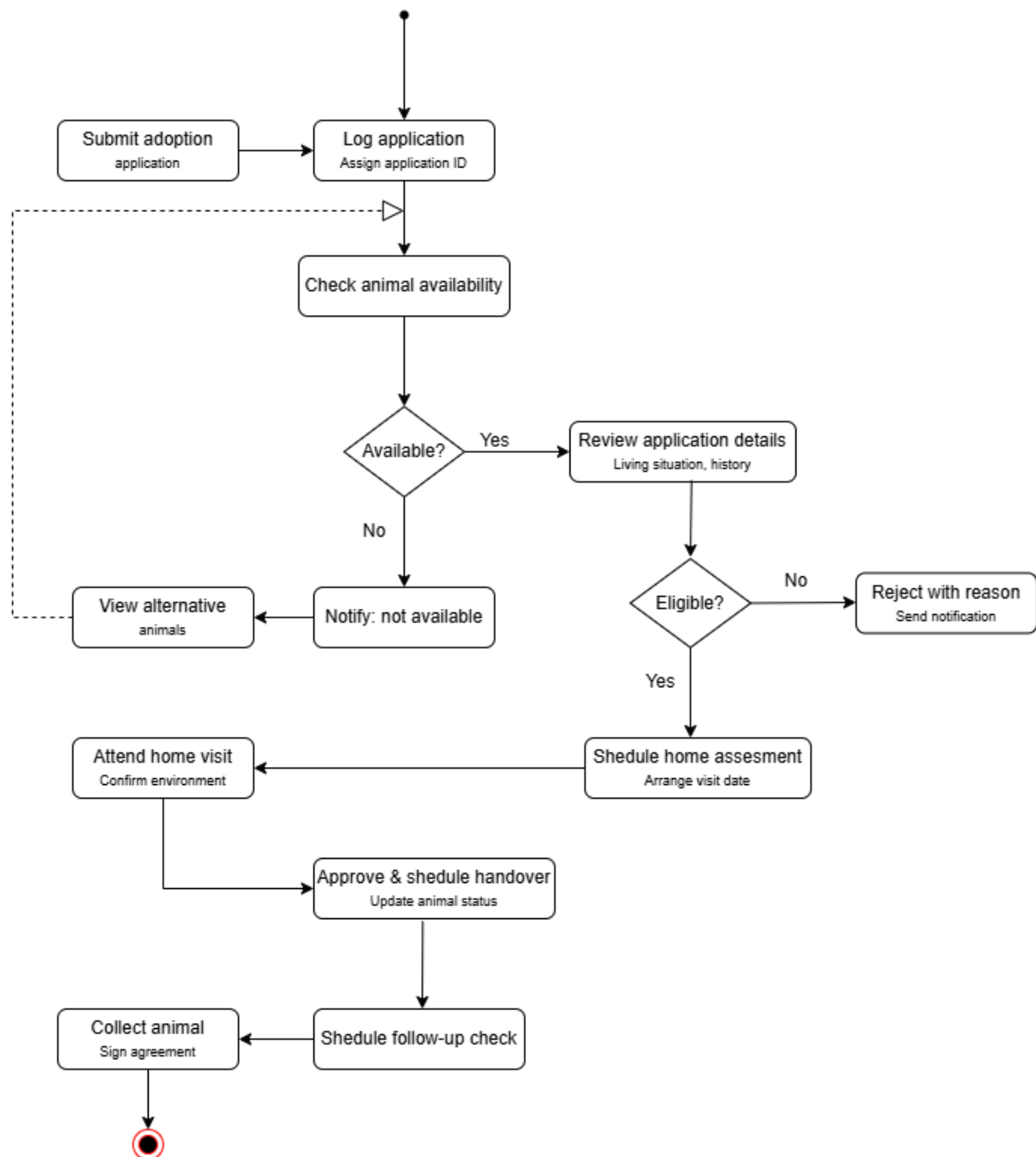
Donation Campaign Flow



Emergency Alert Flow



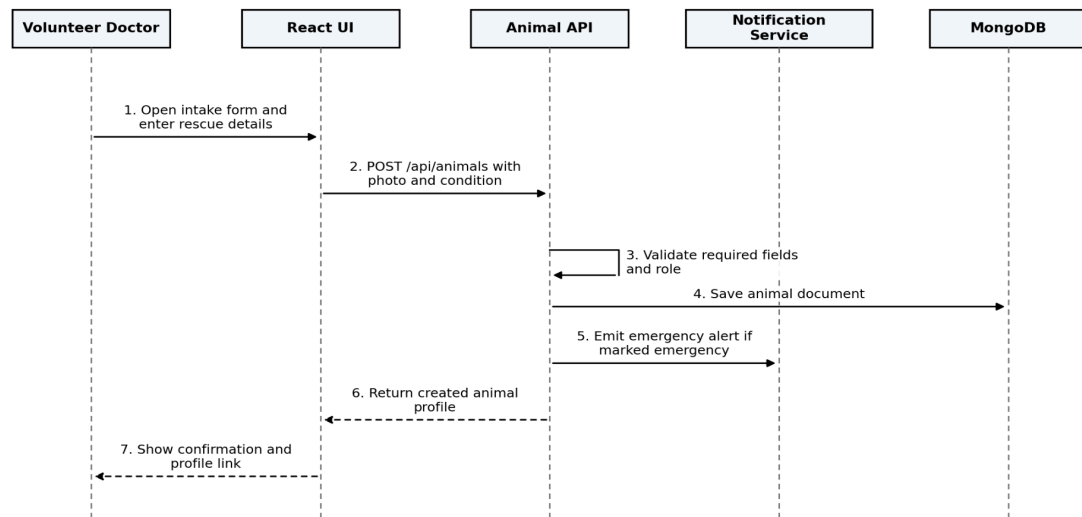
Adoption Process Flow



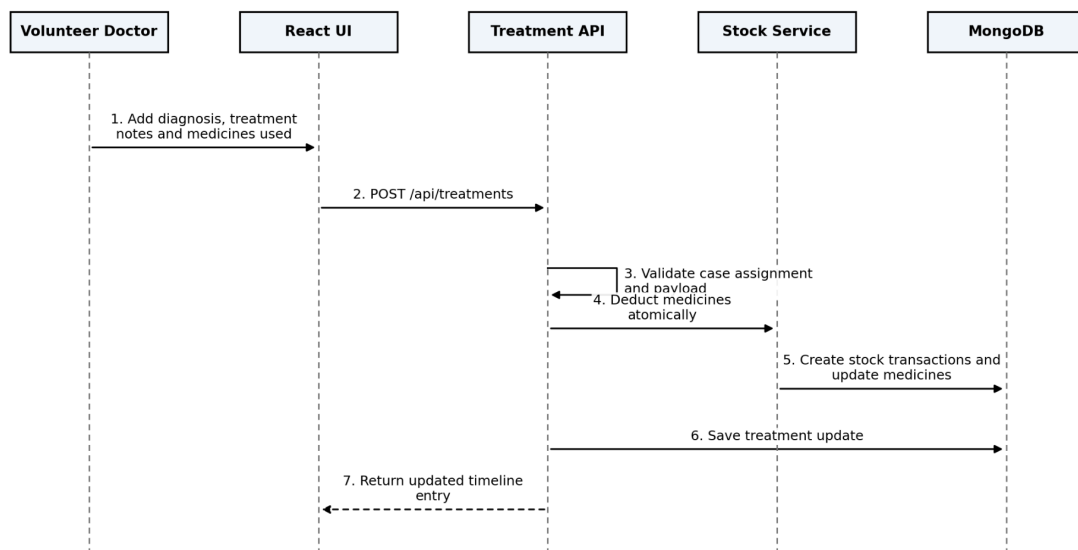
3.7 System Sequence Diagrams

High-level system sequence diagrams should be prepared for the main use cases to show interaction between actors and the Animal Arc system.

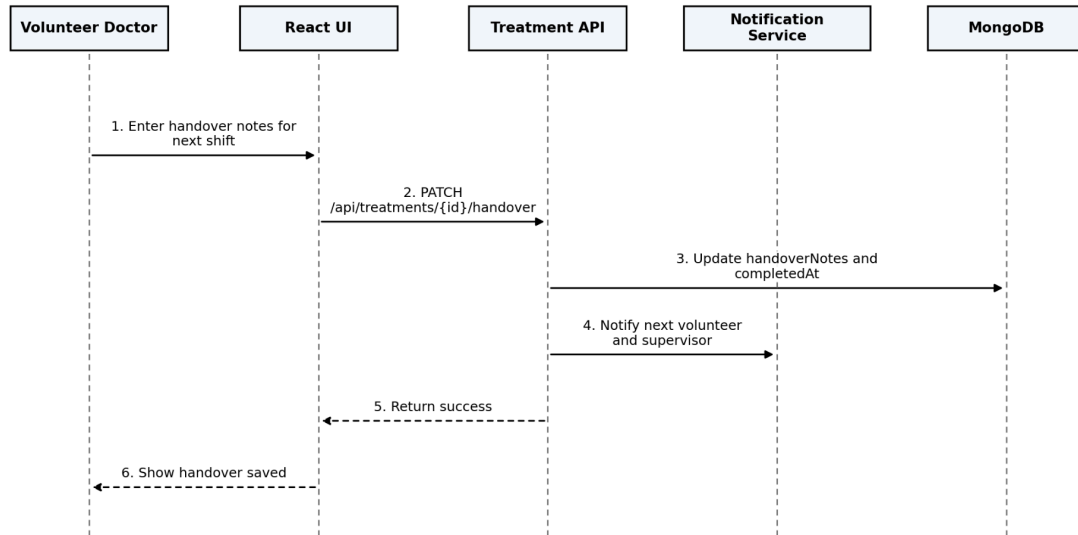
- **SD-01 Animal Intake and Profile Creation [UC-01]**



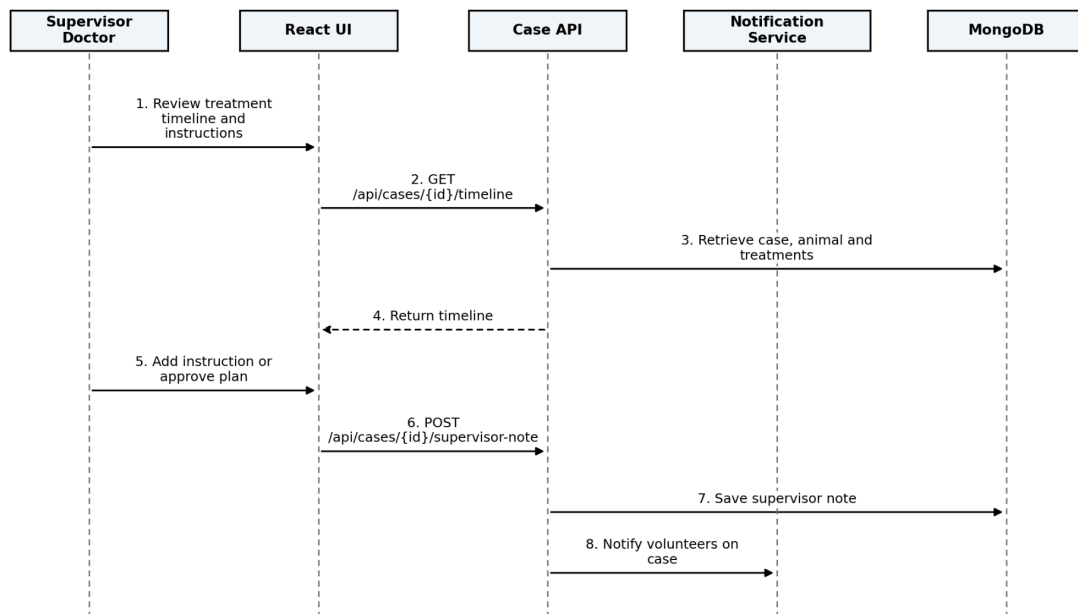
- **SD-02 Treatment Timeline Management [UC-02]**



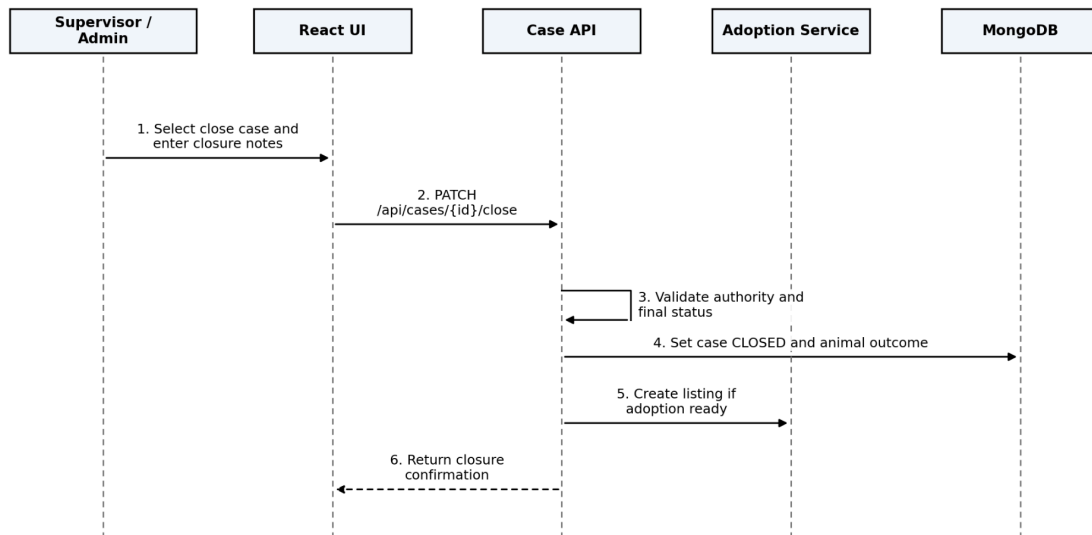
- **SD-03 Handover Notes Management [UC-03]**



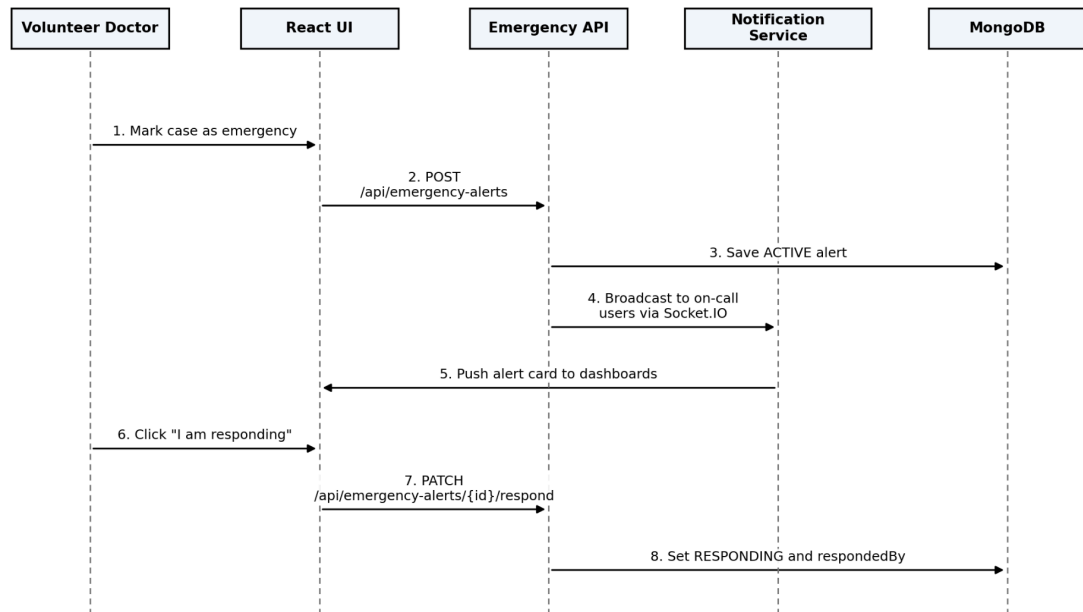
- **SD-04 Supervisor Oversight and Review [UC-04]**



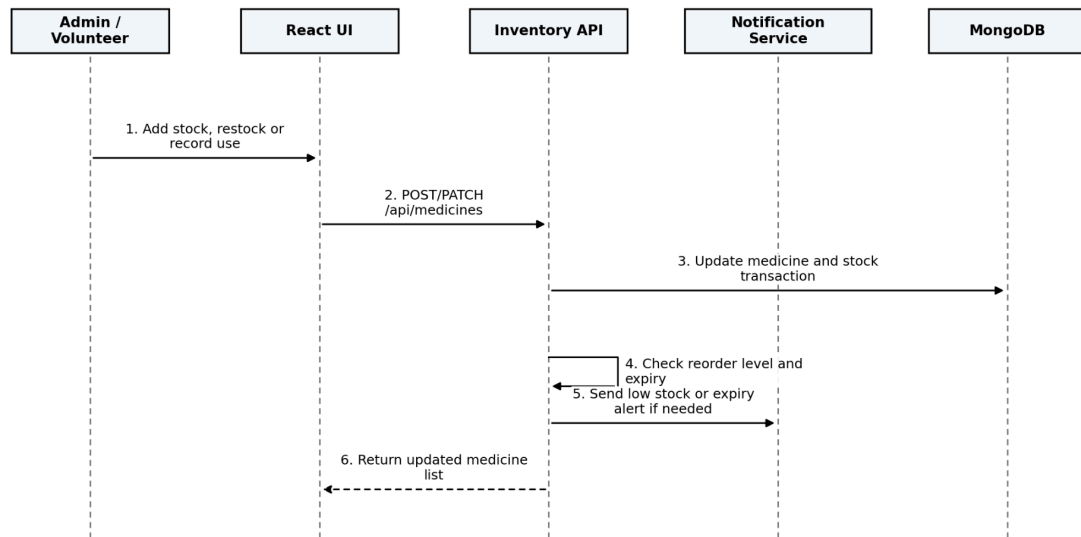
- **SD-05 Case Closure [UC-05]**



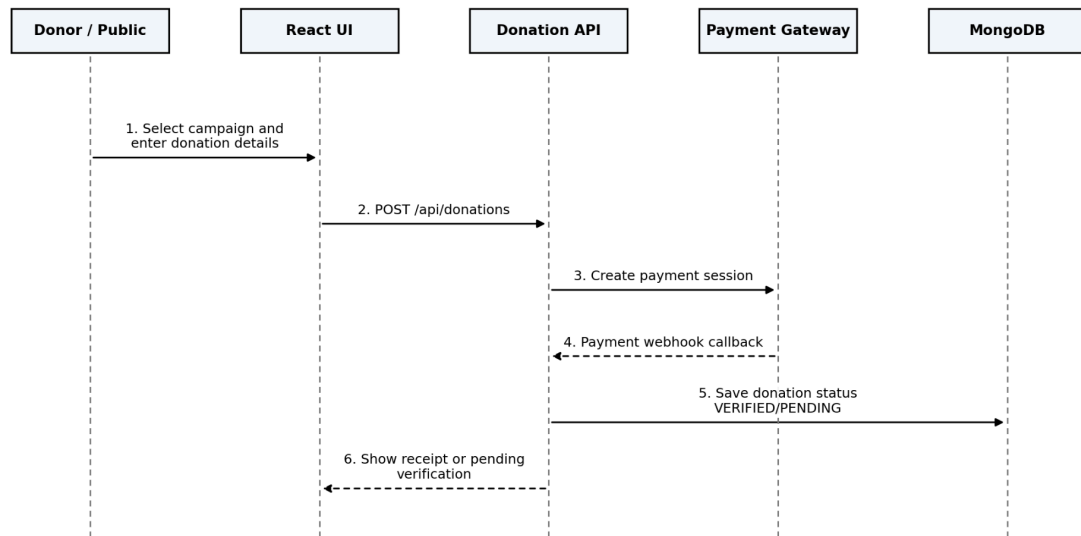
- **SD- 06 Emergency Alert Management [UC-06]**



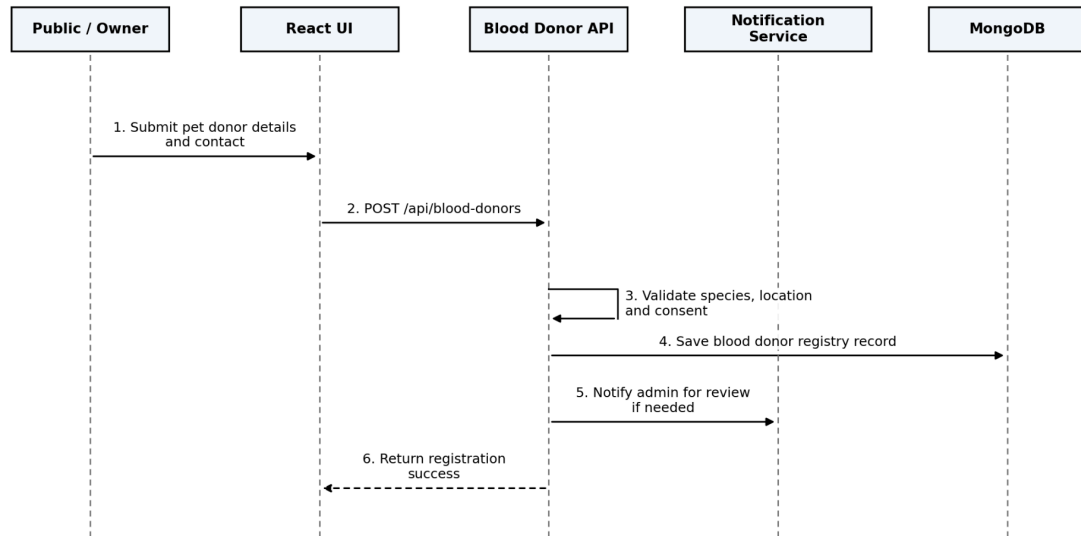
- **SD-07 Medicine Stock Management [UC-07]**



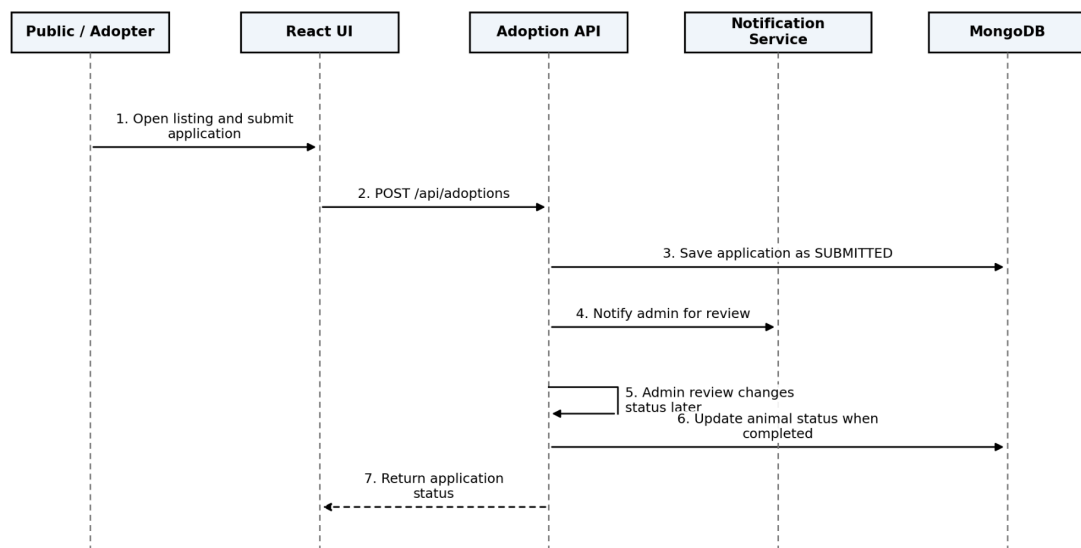
- **SD-08 Donation Campaign and Donation Record Management [UC-08]**



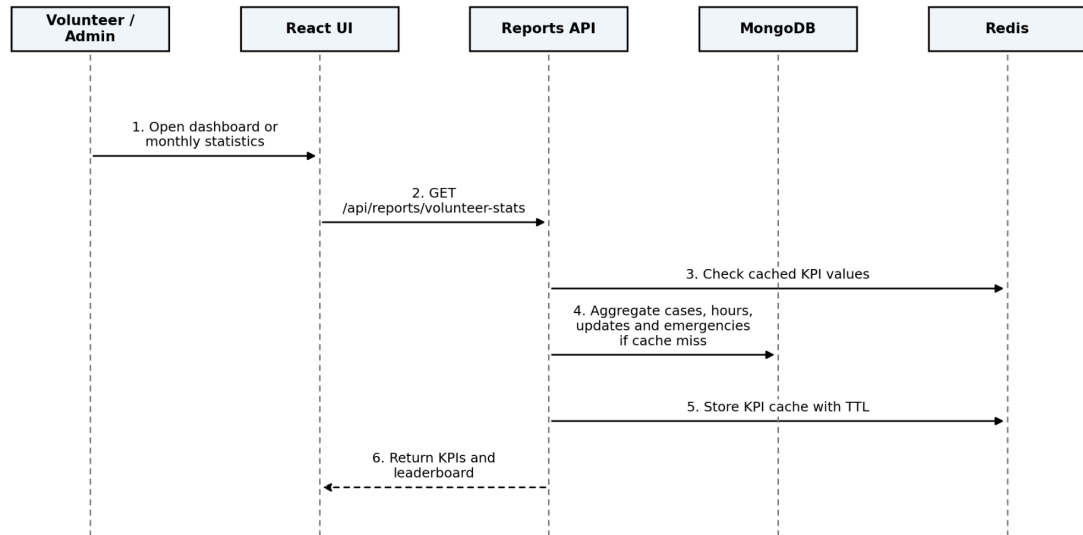
- **SD-09 Blood Donor Registration and Search [UC-09]**



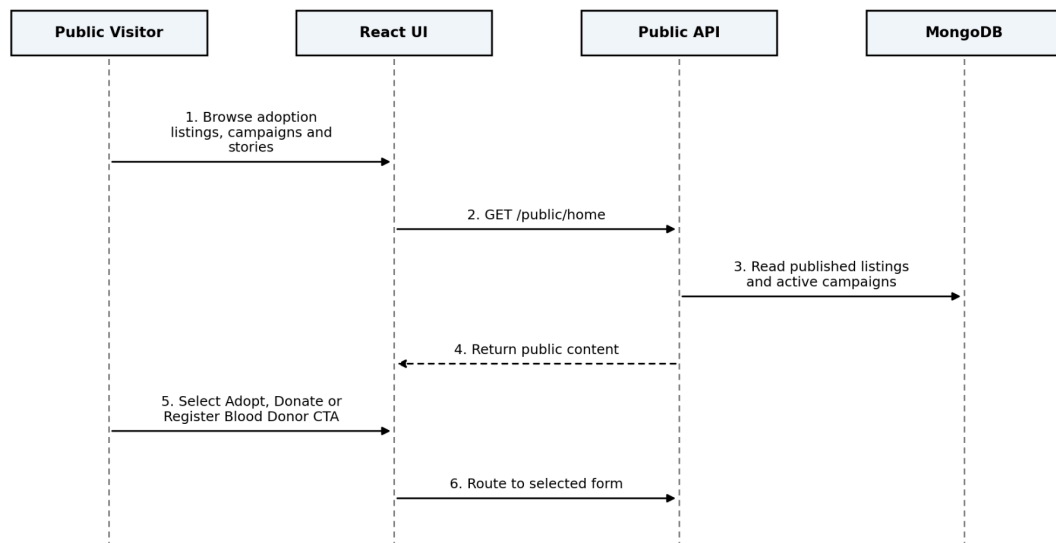
- **SD-10 Adoption Flow Management [UC-10]**



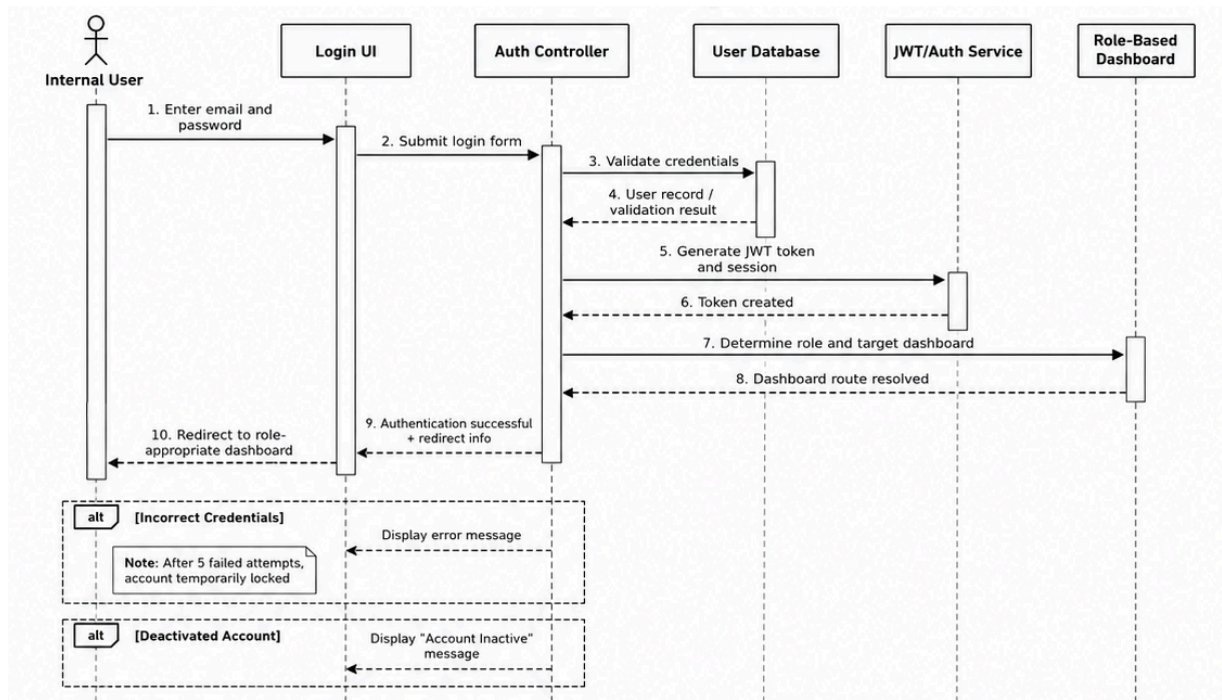
- **SD-11 Volunteer Statistics Tracking [UC-11]**



- **SD-12 Public Portal Access [UC-12]**



- **SD-13 User Authentication and Role Management [UC-13]**



4. Functional Requirements

This section lists the functional requirements of the Animal Arc system. Each requirement is prioritized using MoSCoW prioritization: MUST, SHOULD, and COULD.

FR-01: Animal Intake and Profile Creation

Field	Details
ID	FR-01
Title	Animal Intake and Profile Creation
Description	The system shall allow an authorised volunteer to create a new animal profile by entering rescue details, species, approximate age, sex, condition, rescue location, photos, and initial notes.
Priority	MUST
Source	Stakeholder interview; UC-01 Animal Intake and Profile Creation
Verification Method	Review against UC-01 and test case TC-FR-01

FR-02: Animal Photo and Rescue Location Upload

Field	Details
ID	FR-02
Title	Animal Photo and Rescue Location Upload
Description	The system shall allow an authorised volunteer to upload animal photos and record the rescue location when creating or updating an animal profile.
Priority	SHOULD
Source	Existing System Analysis; UC-01 Animal Intake and Profile Creation
Verification Method	System test TC-FR-02

FR-03: Treatment Timeline Management

Field	Details
ID	FR-03
Title	Treatment Timeline Management
Description	The system shall allow authorised volunteers and supervisors to add treatment updates to an animal profile, including observations, medicines, procedures, treatment dates, and recovery progress.
Priority	MUST
Source	Stakeholder interview; Volunteer Survey Findings; UC-02 Treatment Timeline Management
Verification Method	Review against UC-02 and test case TC-FR-03

FR-04: Handover Notes Management

Field	Details
ID	FR-04
Title	Handover Notes Management
Description	The system shall allow volunteers to add handover notes for active animal cases so that the next volunteer or supervisor can view pending tasks, warnings, feeding instructions, medicine schedules, and important observations.
Priority	MUST
Source	Stakeholder interview; Existing WhatsApp Workflow Analysis; UC-03 Handover Notes Management
Verification Method	Review against UC-03 and test case TC-FR-04

FR-05: Supervisor Oversight and Review

Field	Details
ID	FR-05

Field	Details
Title	Supervisor Oversight and Review
Description	The system shall allow supervisor doctors to view animal profiles, review treatment timelines, add supervisor comments, and approve or reject important case updates where required.
Priority	MUST
Source	Stakeholder interview; UC-04 Supervisor Oversight and Review
Verification Method	Review against UC-04 and test case TC-FR-05

FR-06: Case Closure

Field	Details
ID	FR-06
Title	Case Closure
Description	The system shall allow authorised users to close an animal case by selecting a closure reason such as recovered, adopted, released, transferred, or deceased, and entering final notes.
Priority	MUST
Source	Stakeholder interview; UC-05 Case Closure
Verification Method	Review against UC-05 and test case TC-FR-06

FR-07: Emergency Alert Creation

Field	Details
ID	FR-07
Title	Emergency Alert Creation
Description	The system shall allow authorised volunteers or committee admins to create emergency alerts for urgent rescue or treatment situations by entering the emergency type, location, description, required support, and related animal details if available.
Priority	MUST
Source	Stakeholder interview; Volunteer Survey Findings; UC-06 Emergency Alert Management
Verification Method	Review against UC-06 and test case TC-FR-07

FR-08: Emergency Alert Response Tracking

Field	Details
ID	FR-08
Title	Emergency Alert Response Tracking

Field	Details
Description	The system shall allow authorised users to update the status of an emergency alert as open, assigned, in progress, resolved, or cancelled.
Priority	SHOULD
Source	Existing System Analysis; UC-06 Emergency Alert Management
Verification Method	System test TC-FR-08

FR-09: Medicine Stock Management

Field	Details
ID	FR-09
Title	Medicine Stock Management
Description	The system shall allow authorised committee admins to add, update, and view medicine stock records, including medicine name, quantity, unit, expiry date, storage notes, and last updated date.
Priority	MUST
Source	Stakeholder interview; Existing System Analysis; UC-07 Medicine Stock Management
Verification Method	Review against UC-07 and test case TC-FR-09

FR-10: Low Medicine Stock Alert

Field	Details
ID	FR-10
Title	Low Medicine Stock Alert
Description	The system shall notify authorised users when a medicine or medical supply falls below its defined minimum stock level.
Priority	SHOULD
Source	Volunteer Survey Findings; UC-07 Medicine Stock Management
Verification Method	System test TC-FR-10

FR-11: Donation Campaign Management

Field	Details
ID	FR-11
Title	Donation Campaign Management
Description	The system shall allow committee admins to create and manage donation campaigns by entering campaign title, description, required items or amount, target date, campaign status, and related animal case if applicable.
Priority	MUST
Source	Stakeholder interview; UC-08 Donation Campaign and Donation Record Management

Field	Details
Verification Method	Review against UC-08 and test case TC-FR-11

FR-12: Donation Record Management

Field	Details
ID	FR-12
Title	Donation Record Management
Description	The system shall allow authorised committee admins to record donor details, donation type, donated amount or item, donation date, proof of donation, and related campaign.
Priority	SHOULD
Source	Existing System Analysis; UC-08 Donation Campaign and Donation Record Management
Verification Method	System test TC-FR-12

FR-13: Blood Donor Registration

Field	Details
ID	FR-13
Title	Blood Donor Registration
Description	The system shall allow authorised users or public animal owners to register an animal blood donor by entering owner contact details, animal species, breed, age, weight, blood type if known, location, and availability.
Priority	MUST
Source	Stakeholder interview; UC-09 Blood Donor Registration and Search
Verification Method	Review against UC-09 and test case TC-FR-13

FR-14: Blood Donor Search

Field	Details
ID	FR-14
Title	Blood Donor Search
Description	The system shall allow authorised volunteers and supervisors to search blood donor records by species, location, availability, and other relevant donor details during emergency transfusion situations.
Priority	SHOULD
Source	Stakeholder interview; UC-09 Blood Donor Registration and Search

Field	Details
Verification Method	System test TC-FR-14

FR-15: Adoption Listing Management

Field	Details
ID	FR-15
Title	Adoption Listing Management
Description	The system shall allow authorised users to create and manage adoption listings for animals that are ready for adoption, including animal details, photos, health status, temperament, and adoption requirements.
Priority	MUST
Source	Stakeholder interview; UC-10 Adoption Flow Management
Verification Method	Review against UC-10 and test case TC-FR-15

FR-16: Adoption Application Management

Field	Details
ID	FR-16
Title	Adoption Application Management
Description	The system shall allow public users to submit adoption applications and allow authorised committee admins to review, approve, reject, or mark applications for follow-up.
Priority	MUST
Source	Stakeholder interview; UC-10 Adoption Flow Management
Verification Method	Review against UC-10 and test case TC-FR-16

FR-17: Volunteer Statistics Tracking

Field	Details
ID	FR-17
Title	Volunteer Statistics Tracking
Description	The system shall calculate and display volunteer contribution statistics such as number of cases handled, treatment updates added, emergency responses, handover notes, and adoption support activities.
Priority	MUST
Source	Volunteer Survey Findings; UC-11 Volunteer Statistics Tracking
Verification Method	Review against UC-11 and test case TC-FR-17

FR-18: Public Portal

Field	Details
ID	FR-18
Title	Public Portal
Description	The system shall provide a public portal where visitors can view adoption listings, donation campaigns, blood donor registration information, awareness content, and Animal Ark contact details.
Priority	MUST
Source	Stakeholder interview; Existing System Analysis; UC-12 Public Portal Access
Verification Method	Review against UC-12 and test case TC-FR-18

FR-19: Public Inquiry Submission

Field	Details
ID	FR-19
Title	Public Inquiry Submission
Description	The system shall allow public users to submit inquiries related to adoption, donations, animal rescue, blood donation, or general support.
Priority	SHOULD
Source	Public Portal Requirement; UC-12 Public Portal Access
Verification Method	System test TC-FR-19

FR-20: User Authentication and Role Management

Field	Details
ID	FR-20
Title	User Authentication and Role Management
Description	The system shall allow registered internal users to log in securely and shall enforce role-based access control for volunteers, supervisor doctors, committee admins, and public users.
Priority	MUST
Source	System Security Requirement; UC-13 User Authentication and Role Management
Verification Method	Security test TC-FR-20

5. Non-Functional Requirements

This section defines the quality attributes and constraints of the Animal Ark system.

5.1 Performance

- NFR-PER-01: The system shall load the main dashboard within 3 seconds for 95% of requests under normal operating conditions.
- NFR-PER-02: The system shall save a completed animal intake record within 2 seconds after the volunteer submits the form.
- The system shall optimize uploaded images to reduce storage usage and improve loading performance.
- NFR-PER-04: The system shall retrieve and display treatment history containing up to 500 treatment records within 2 seconds.
- NFR-PER-05: The system shall support at least 100 concurrent users without degradation of core functionality.
- NFR-PER-06: Emergency alert notifications shall be delivered to relevant users within 10 seconds of alert creation.

5.2 Security

- NFR-SEC-01: The system shall enforce role-based access control for volunteers, supervisor doctors, committee admins, donors, adopters, and public visitors.
- NFR-SEC-02: Public users shall not be able to view internal treatment records, volunteer notes, medicine stock details, or private donor/adopter information.
- NFR-SEC-03: User passwords shall be stored securely using accepted password hashing techniques.
- NFR-SEC-04: The system shall protect sensitive contact details of donors, adopters, blood donor owners, and volunteers.
- NFR-SEC-05: The system shall require authentication before allowing users to create, update, or delete internal records.

5.3 Usability

- NFR-USA-01: The system interface shall be simple and easy to use for volunteer veterinary students who may need to update records during busy treatment or rescue activities.
- NFR-USA-02: The system shall use clear labels, simple forms, and meaningful status indicators for animal cases, emergency alerts, donations, and adoption applications.
- NFR-USA-03: The public portal shall be understandable for users who are not familiar with Animal Ark.
- NFR-USA-04: The system shall provide mobile-responsive pages for volunteers and public users accessing the system through smartphones.

5.4 Reliability

- NFR-REL-01: The system shall preserve submitted animal records, treatment updates, donation records, and adoption applications without data loss under normal operating conditions.
- NFR-REL-02: The system shall validate required fields before saving important records such as animal intake, treatment updates, emergency alerts, and adoption applications.
- NFR-REL-03: The system shall display appropriate error messages when records cannot be saved or loaded.
- NFR-REL-04: The system shall maintain accurate status values for animal cases, emergency alerts, donation campaigns, and adoption applications.

5.5 Scalability

- NFR-SCA-01: The system database design shall support growth in animal records, treatment updates, photos, donation records, adoption applications, and volunteer activity logs.
- NFR-SCA-02: The public portal shall support multiple public users viewing adoption and donation information at the same time.
- NFR-SCA-03: The system architecture shall allow additional modules to be added in future, such as advanced reporting, payment integration, or mobile app support.

5.6 Maintainability

- NFR-MNT-01: The system shall be designed using modular components so that animal intake, treatment, donations, adoption, medicine stock, and public portal features can be maintained separately.
- NFR-MNT-02: The system shall use consistent naming conventions for database tables, API endpoints, and user interface components.
- NFR-MNT-03: The system shall maintain clear documentation for use cases, functional requirements, diagrams, and test cases.
- NFR-MNT-04: The system shall allow future developers to update the system without needing to rewrite unrelated modules.

6. Alternative Solutions and Feasibility Study

6.1 Alternative Solutions Analysed

Before recommending the Animal Arc system, the following alternative solutions were analysed.

Alternative 1: Continue Using WhatsApp, Paper Records, and Manual Coordination

Description: Animal Ark could continue managing animal cases, treatment updates, donations, adoption posts, and emergency communication through WhatsApp messages, handwritten notes, spreadsheets, and informal verbal communication.

Pros:

- No new system development cost.
- Volunteers are already familiar with WhatsApp and manual communication.
- Easy to continue immediately without training.

Cons:

- Treatment records remain scattered and difficult to search.
- Handover details may be missed between volunteers.
- Emergency communication may not reach all responsible users quickly.
- Medicine stock records may be inaccurate or outdated.
- Donation and adoption records may be difficult to track.
- Volunteer contribution statistics cannot be generated reliably.
- Long-term animal history may be lost.

Alternative 2: Use Generic Tools such as Google Forms, Google Sheets, and Social Media

Description: Animal Ark could use general-purpose tools such as Google Forms for data collection, Google Sheets for tracking records, and social media pages for adoption and donation communication.

Pros:

- Low cost and easy to set up.
- Familiar tools for many students.
- Useful for basic form submissions and simple records.

Cons:

- Does not provide a proper animal treatment timeline.
- Does not support structured supervisor oversight.
- Does not provide integrated medicine stock, adoption, donation, and blood donor workflows.
- Difficult to enforce role-based access properly.
- Data may become duplicated across multiple files.
- Not suitable as a long-term centralized animal welfare management system.

Alternative 3: Develop the Custom Animal Ark System

Description: A custom software system can be designed specifically for Animal Ark rescue, treatment, donation, adoption, volunteer, medicine stock, and public engagement workflows.

Pros:

- Directly matches Animal Ark operational needs.
- Centralizes animal profiles and treatment history.
- Improves handover communication and emergency response.
- Supports medicine stock awareness.
- Provides structured donation and adoption management.
- Enables volunteer statistics and reporting.
- Provides a public portal for donors, adopters, and supporters.
- Can be extended in the future.

Cons:

- Requires design, development, testing, and deployment effort.
- Requires user training and adoption.
- Requires ongoing maintenance after deployment.

6.2 Feasibility Study

6.2.1 Technical Feasibility

The proposed Animal Ark system is technically feasible. The required features such as user authentication, role-based access control, animal profiles, treatment timelines, image uploads, stock management, donation records, adoption applications, and public portal pages can be implemented using common web development technologies. The system does not require highly complex technologies such as artificial intelligence, advanced medical devices, or real-time GPS tracking in the initial version. Therefore, the technical risk is manageable within the project scope.

6.2.2 Economic Feasibility

The system is economically feasible because it can be developed as an academic project using open-source technologies. The main costs would be related to hosting, storage, and maintenance if the system is deployed for real use. Compared to commercial animal shelter or veterinary management systems, a custom academic solution is more suitable because it can focus only on Animal Ark actual needs without unnecessary paid features.

6.2.3 Operational Feasibility

The system is operationally feasible because it directly addresses current problems faced by Animal Ark volunteers and committee members. The system supports existing workflows such as rescue intake, treatment updates, handovers, emergency coordination, donation management, and adoption handling. Since the system is designed around current Animal Ark activities, users are more likely to adopt it if the interface is simple, mobile-friendly, and useful during real volunteer work.

6.2.4 Schedule Feasibility

The system is feasible within the academic project timeline if the scope is controlled properly. The SRS and system design can be completed during the SENG 31242 period, while implementation can focus on the most important modules first.

The highest priority modules should be:

1. Animal intake and profile creation.
2. Treatment timeline.
3. Handover notes.
4. Emergency alerts.
5. Medicine stock.
6. Adoption flow.
7. Public portal.

Lower-priority enhancements such as advanced analytics, online payments, and mobile app development can be considered in future versions.

6.3 Justified Recommendation

Based on the analysis, the recommended solution is to develop the custom Animal Arc system.

Continuing with WhatsApp, paper records, and manual coordination does not solve the core problems of scattered treatment records, weak handovers, stock uncertainty, donation tracking difficulties, and unstructured adoption management. Generic tools such as Google Forms and Google Sheets can support simple record keeping, but they do not provide an integrated

workflow for animal rescue, treatment, emergency handling, donations, adoptions, and volunteer statistics.

Animal Arc is the most suitable solution because it is designed specifically for Animal Ark operational needs. It provides a centralized, role-based, and scalable platform that can improve animal care continuity, volunteer coordination, public engagement, and long-term record management.